STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 13: Oscillations

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

Time for one cycle.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q2 1 mark

f=1/T.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q3 1 mark

F=-kx.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q4 1 mark

At equilibrium.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q5 1 mark

At extreme displacement.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q6 1 mark

sqrt(k/m).

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q7 1 mark

2pisqrt(l/g).

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q8 1 mark

kA^2.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q9 2 marks

2 Hz.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q10 2 marks

0.314 s.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q11 2 marks

a=-omega^2x.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q12 2 marks

2.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q13 2 marks

Zero.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q14 2 marks

Small angular displacement.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q15 3 marks

omega=20 rad/s; T~0.314 s.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q16 3 marks

0.4 m/s and 1.6 m/s^2.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q17 3 marks

~2.01 s.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q18 3 marks

K max/U min at equilibrium; U max/K zero at extremes; total energy constant without damping.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q19 4 marks

E=1 J; U=.25 J.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.

Q20 4 marks

Damping transfers mechanical energy into heat/sound and reduces amplitude.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.